

Master Circuit Breaker Value vs. Measured Values Analysis



[UC13-01] - Awesense Energy Transition Platform Use Case

The Utility Problem

The master circuit breaker value is very important for every electricity customer. In many electricity distribution utilities, this value represents a maximum capacity (maximum load) that cannot be exceeded by the consumer at any given time. However, this value is very often oversized, either historically, when the customer had no idea of their consumption, or because the customer does not compare this value (maximum capacity) with their real measured time-series data or for other reasons.

These unnecessarily-high master circuit breaker values impact both consumers and electric utilities. On the customer side, they are paying unnecessarily high charges to reserve this capacity (capacity charges). On the utility side, these contracted capacities are unnecessarily blocked, and when capacity is lacking in the area, the utility has to undertake costly grid upgrades. These upgrades are then usually paid by all service area customers as a collective payment.

Having the values of the master circuit breaker adjusted compared to actual consumption values brings multiple opportunities. First, customer satisfaction and customer engagement, where the distribution companies can provide information to customers and if they can reduce this parameter, they will save monthly payments for booking lower capacity. However, these adjustments can also be used directly for the utility in Grid Planning. Based on the value of the master circuit breaker, the grid available capacity is usually calculated. Therefore, if the master circuit breaker values are adjusted according to actual measured values, there would be no need to reinforce the grid (e.g. by replacing a transformer or conductors) based on insufficient capacity in the area.

The Data Solutions and Methods

For conducting such an analysis, it is necessary to have two datasets ready. These datasets are as follows:

1. Consumption time series from the utility meter (consumption point).
2. Master circuit breaker value of consumption point.

Building such analysis using the Awesense Energy Transition Platform is easy. To get the necessary datasets for this analysis, the Awesense Energy Data Model (EDM) interface offers these methods:

1. Consumption time series from the utility meter (consumption point):
 - a. Querying time series values is very simple in EDM. The maximum value can be found by:
 - i. First: querying the time series,
 - ii. Second: finding the maximum of the queried time series.

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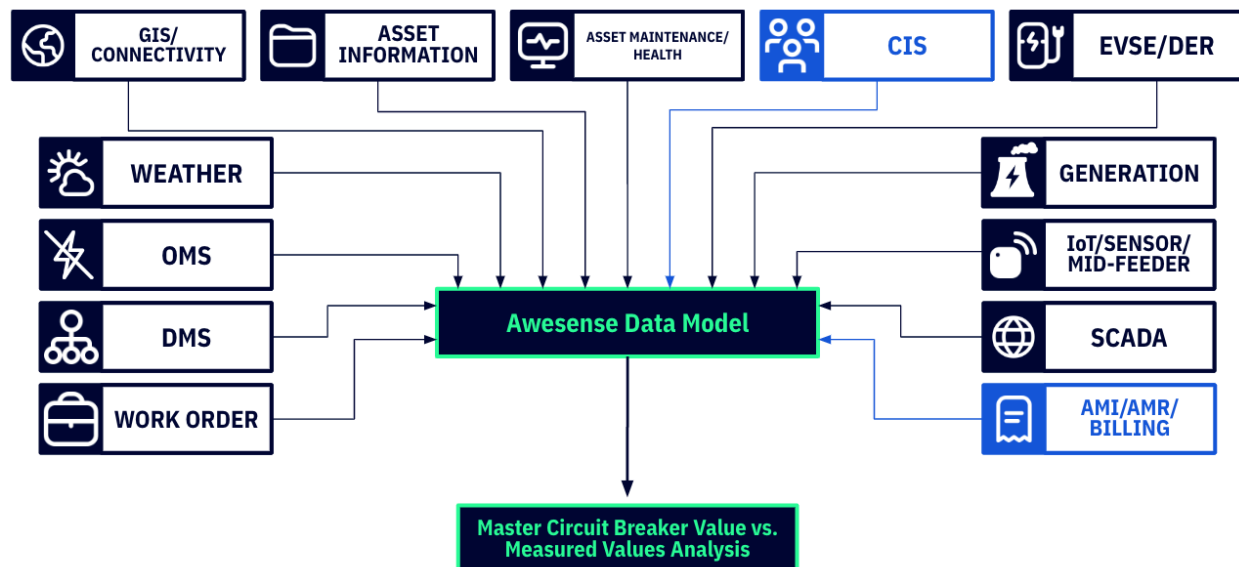
2. Master circuit breaker value of consumption point:
 - a. Information about the capacity of consumption point is included in the EDM model and is therefore very easy to get. This value is stored as an attribute of the meter element.

The analyst would then create a simple calculation which compares the above values for each consumption point (maximum measured value from the utility meter time series and the master circuit breaker value). The results might then be exported for all consumption points and ranked according to the most significant percentage difference.

These results can be further used for contacting customers. Customer contact details can be stored directly (again) in EDM, and these details will also be included in the export. These results can also be further used in Grid Planning (to find areas with significant differences). Individual points might be shown on the map by the size of the point based on the results (the larger the points, the greater the free capacity at the consumption point).

Required Data Sets

The required data for this use case are visualized in the following diagram. Data sources highlighted in [royal blue](#) are required for this use case.



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Available Use Case Implementations:

Users of the Awesense Energy Transition Platform can access this use case by the following means:

1. GitHub [link](#) to Jupyter notebook.